

Project 07: Deep Research Intelligence Platform

Autonomous Multi-Step AI Research Agent

1. Executive Overview

The Deep Research Intelligence Platform executes multi-step research workflows. It decomposes user research topics into targeted sub-questions, gathers evidence from an indexed corpus, synthesizes structured technical reports with inline citations, reflects on missing knowledge gaps, and revises final outputs.

2. System Architecture & Workflow Pipeline

- Step 1: Decompose Topic into Sub-Questions
- Step 2: Gather Evidence from Indexed Web Corpus
- Step 3: Synthesize Grounded Report Draft with Citations
- Step 4: Self-Reflection & Gap Detection Scoring
- Step 5: Revise & Finalize Comprehensive Report

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves Deep Research Platform web UI
POST	/api/research	Executes 5-step research pipeline and returns report & evidence

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 07_deep_research/src && python -m uvicorn server:app --host 127.0.0.1 --port 8007
```

Automated Test Suite Execution:

```
python -m pytest 07_deep_research/tests/
```